# Proof by Contradiction and an Introduction to Functions

**Solutions**

## Part I. Proof by contradiction — 56 minutes (70%)

For each problem, write a proof by contradiction. State your assumption, identify the contradiction explicitly, and conclude the original claim.

**Recall.** To prove a statement $P$ by contradiction, assume $\neg P$ and derive a contradiction. To prove $P \Rightarrow Q$ this way, assume *both* $P$ and $\neg Q$.

### 1. Sets

Let $A$ and $B$ be sets. Prove that

$$A \cap (B \setminus A) = \emptyset.$$

If the intersection were nonempty, what element could you choose? Explain which two membership statements would contradict each other.

**Solution.** Suppose, for a contradiction, that $A \cap (B \setminus A) \neq \emptyset$. Choose $x \in A \cap (B \setminus A)$. Then $x \in A$ and $x \in B \setminus A$. The latter means that $x \in B$ and $x \notin A$. Thus $x \in A$ and $x \notin A$, a contradiction. Therefore $A \cap (B \setminus A) = \emptyset$.

### 2. There is no largest even integer

Prove that there is no largest even integer.

Begin by stating precisely what it would mean for an even integer $m$ to be the largest. Construct an even integer that contradicts that assumption.

**Solution.** Suppose, for a contradiction, that a largest even integer $m$ exists. Thus every even integer is at most $m$. Since $m$ is even, $m = 2k$ for some $k \in \mathbb{Z}$. But

$$m + 2 = 2k + 2 = 2(k + 1)$$

is also even, because $k + 1 \in \mathbb{Z}$, and $m + 2 > m$. This contradicts the maximality of $m$. Hence there is no largest even integer.

## Part I. Proof by contradiction (continued)

### 3. Even squares

Let $a \in \mathbb{Z}$. Prove that if $a^2$ is even, then $a$ is even.

First write down the hypothesis together with the negation of the conclusion. You may use the fact that every integer is either even or odd, but not both.

**Solution.** Suppose that $a \in \mathbb{Z}$, that $a^2$ is even, and, for a contradiction, that $a$ is not even. Then $a$ is odd, so $a = 2k + 1$ for some $k \in \mathbb{Z}$. Consequently,

$$a^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k \in \mathbb{Z}$, this shows that $a^2$ is odd. This contradicts the hypothesis that $a^2$ is even. Therefore $a$ is even.

### 4. Rational and irrational numbers

Let $a, b \in \mathbb{R}$. Prove that if $a$ is rational and $ab$ is irrational, then $b$ is irrational.

Use the definition of a rational number. When you introduce fractions, state which quantities are integers and which denominators are nonzero.

**Solution.** Suppose that $a \in \mathbb{Q}$ and $ab \notin \mathbb{Q}$. For a contradiction, suppose that $b \in \mathbb{Q}$. There exist $k, \ell, m, n \in \mathbb{Z}$, with $m \neq 0$ and $n \neq 0$, such that

$$a = \frac{k}{m}, \qquad b = \frac{\ell}{n}.$$

Then $ab = \dfrac{k\ell}{mn}$. The numerator $k\ell$ is an integer and the denominator $mn$ is a nonzero integer. Thus $ab \in \mathbb{Q}$, contradicting $ab \notin \mathbb{Q}$. Therefore $b$ is irrational.

**Check your proof.** Underline the assumption you made for contradiction and circle the two statements that cannot both be true.

In Problem 4, the additional assumption is $b \in \mathbb{Q}$; the contradictory statements are $ab \in \mathbb{Q}$ and $ab \notin \mathbb{Q}$.

## Part I. Putting the arguments together

### 5. An impossible equation

Prove that for every $a, b \in \mathbb{Z}$,

$$a^2 - 4b \neq 2.$$

You may use the result of Problem 3. If the equation $a^2 - 4b = 2$ held, what would it tell you about the parity of $a^2$, and then of $a$?

**Solution.** Suppose, for a contradiction, that there exist $a, b \in \mathbb{Z}$ with $a^2 - 4b = 2$. Then

$$a^2 = 4b + 2 = 2(2b + 1),$$

so $a^2$ is even. By Problem 3, $a$ is even; write $a = 2k$ with $k \in \mathbb{Z}$. Substitution gives

$$4k^2 - 4b = 2, \qquad \text{hence} \qquad 2(k^2 - b) = 1.$$

Since $k^2 - b \in \mathbb{Z}$, this would make 1 even, a contradiction. Therefore $a^2 - 4b \neq 2$ for all $a, b \in \mathbb{Z}$.

### 6. A sum of two squares

Let $a, b, c \in \mathbb{Z}$. Prove that if $a^2 + b^2 = c^2$, then at least one of $a$ or $b$ is even.

Negate "at least one is even" carefully. You may use Problems 3 and 5.

**Solution.** Suppose that $a, b, c \in \mathbb{Z}$ satisfy $a^2 + b^2 = c^2$. For a contradiction, suppose that neither $a$ nor $b$ is even. Then both are odd, so write $a = 2k + 1$ and $b = 2m + 1$, where $k, m \in \mathbb{Z}$. We obtain

$$\begin{aligned} c^2 &= (2k + 1)^2 + (2m + 1)^2 \\ &= 4(k^2 + k + m^2 + m) + 2 \\ &= 2\big(2(k^2 + k + m^2 + m) + 1\big). \end{aligned}$$

Set $s = 2(k^2 + k + m^2 + m) + 1$, which is an odd integer. Since $c^2 = 2s$ is even, Problem 3 implies that $c$ is even. Write $c = 2n$ with $n \in \mathbb{Z}$. Then $4n^2 = 2s$, so $s = 2n^2$ is even. This contradicts the fact that $s$ is odd. Therefore at least one of $a$ or $b$ is even.

**Before moving on.** In a proof of $P \Rightarrow Q$ by contradiction, why must you retain the hypothesis $P$ when you assume $\neg Q$?

The negation of $P \Rightarrow Q$ is $P \wedge \neg Q$. An implication fails only when its hypothesis is true and its conclusion is false. Assuming only $\neg Q$ does not describe a counterexample to the implication; we must also assume $P$.

## Part II. Functions 24 minutes (30%)

**Definitions.** A function $f\colon X \to Y$ assigns to each $x \in X$ exactly one value $f(x) \in Y$. Its **domain** is $X$, its **codomain** is $Y$, and its **image** is $f(X) = \{f(x) : x \in X\}$.

- **Injective (one-to-one):** for all $x_1, x_2 \in X$, $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$.
- **Surjective (onto):** for every $y \in Y$, there is an $x \in X$ with $f(x) = y$.

### 7. The same formula with different domains and codomains

For each assignment, decide whether it defines a function. If it does, give its image and decide whether it is injective and whether it is surjective. If it does not, give an input that shows the failure and mark the last three columns N/A. Give brief explanations below the table.

| Assignment | Function? | Image | Injective? | Surjective? |
|---|---|---|---|---|
| $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x/5$ | Yes | $\mathbb{R}$ | Yes | Yes |
| $g\colon \mathbb{Z} \to \mathbb{Z}$, $g(x) = x/5$ | No | N/A | N/A | N/A |
| $h\colon \mathbb{Q} \to \mathbb{R}$, $h(x) = x/5$ | Yes | $\mathbb{Q}$ | Yes | No |
| $\ell\colon \mathbb{R} \to \mathbb{Q}$, $\ell(x) = x/5$ | No | N/A | N/A | N/A |

**Explanations:** For an injective function, explain why equal outputs force equal inputs. For a surjective function, show how to find an input for an arbitrary target; for a nonsurjective function, identify a target outside its image.

**Solution.** **(a)** Every real $x$ gives exactly one real output $x/5$, so $f$ is a function. If $f(x_1) = f(x_2)$, then $x_1/5 = x_2/5$, hence $x_1 = x_2$; thus $f$ is injective. Given any $y \in \mathbb{R}$, the input $x = 5y \in \mathbb{R}$ satisfies $f(x) = y$. Thus its image is $\mathbb{R}$ and it is surjective.

**(b)** This is not a function $\mathbb{Z} \to \mathbb{Z}$: the input $2 \in \mathbb{Z}$ gives $2/5 \notin \mathbb{Z}$.

**(c)** Every rational $x$ gives a unique rational (hence real) output $x/5$, so $h$ is a function. The same equality argument as in (a) proves injectivity. All outputs lie in $\mathbb{Q}$, and every $q \in \mathbb{Q}$ is attained at $5q \in \mathbb{Q}$. Hence $h(\mathbb{Q}) = \mathbb{Q}$. It is not surjective onto $\mathbb{R}$, since, for example, $\sqrt{2} \notin \mathbb{Q}$ is not in its image.

**(d)** This is not a function $\mathbb{R} \to \mathbb{Q}$. The input $\sqrt{2}$ would give $\sqrt{2}/5 \notin \mathbb{Q}$: if that quotient were rational, its product with 5 would make $\sqrt{2}$ rational.

**Discuss.** How can the formula $x/5$ define a function in one row but fail to define a function in another? What is the difference between the image and the codomain?

A formula defines a function $X \to Y$ only if every input in $X$ has exactly one output in $Y$. Changing either set can change whether this condition holds. The codomain is the specified target set; the image consists of the outputs actually attained. For example, $h\colon \mathbb{Q} \to \mathbb{R}$ has codomain $\mathbb{R}$ but image $\mathbb{Q}$.

## Part II. Injectivity and surjectivity (continued)

### 8. Disproving injectivity

Prove or disprove: $f\colon \mathbb{R} \to \mathbb{R}$, defined by $f(x) = x^3 - x$, is injective. If you disprove it, exhibit two distinct inputs with the same output.

**Solution.** The statement is false. The inputs 1 and 0 are distinct, but

$$f(1) = 1^3 - 1 = 0 = 0^3 - 0 = f(0).$$

Thus distinct inputs have the same output, so $f$ is not injective.

### 9. Proving surjectivity

Prove that $g\colon \mathbb{R}^2 \to \mathbb{R}^2$, defined by $g(x, y) = (x + y, x - y)$, is surjective.

Start with an arbitrary target $(u, v) \in \mathbb{R}^2$. Find real numbers $x, y$ satisfying $x + y = u$ and $x - y = v$, and check your answer.

**Solution.** Let $(u, v) \in \mathbb{R}^2$ be arbitrary. Solving $x + y = u$ and $x - y = v$ suggests choosing

$$x = \frac{u + v}{2}, \qquad y = \frac{u - v}{2}.$$

Both $x$ and $y$ are real, so $(x, y) \in \mathbb{R}^2$. Moreover,

$$\begin{aligned} g(x, y) &= \left( \frac{u + v}{2} + \frac{u - v}{2}, \frac{u + v}{2} - \frac{u - v}{2} \right) \\ &= (u, v). \end{aligned}$$

Thus every target in $\mathbb{R}^2$ has a preimage in $\mathbb{R}^2$, and $g$ is surjective.